INCH-POUND

MIL-C-50388A (AR)
26 JUNE 1989
SUPERSEDING
MIL-C-50388 (MU)
6 July 1970

MILITARY SPECIFICATION

CARTRIDGE, 20MM, HIGH EXPLOSIVE INCENDIARY, M210

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and preparation for delivery for Cartridge, 20mm, High Explosive Incendiary, M210 for use in the M3 and MK16 20mm Automatic Weapon Systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking for Small Arms Ammunition
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

7259064 - Projectile, 20mm, HEI for the M210 Cartridge
7259065 - Charged Body, Projectile, 20mm, HEI for the M210 Cartridge
7259105 - Cartridge, 20mm, High Explosive Incendiary, M210
7553663 - Classification of Firing Casualties for: Case, Cartridge, 20mm, M21A1 and M21A1B1

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INSPECTION EQUIPMENT DRAWINGS

7259443	- Rack Cartridge
7259469	- Torque, Test Gage
7286068	- Limit Check Taper Plug Gage
7286069	- Limit Check Taper Plug Gage
7286070	- Limit Check Plug Gage
7276071	- Limit Check Plug Gage
7299942	- Chamber Gage
7556901	- Indicator, Comparator Gage
7569084	- Indicator Gage
8655592	- Ring Gage
11075013	- Gage, Feeler
11075131	- Fixture, Test

PUBLICATIONS

US. ARMY MATERIAL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedure for 20mm Cartridges

CODE OF FEDERAL REGULATIONS

49 CFR-100-199 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Copies of the Code of Federal Regulations are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify "49 CFR-100-199 (Latest Revision)".)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however shall supersede applicable laws and regulations unless specific exemption has been obtained.

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3 REQUIREMENTS

3.1 General.

a. Material. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.6).

3.2 Cartridge. The cartridge shall comply with all the requirements specified on Drawing (Dwg.) 7259105, all associated drawings and with all requirements specified in applicable specifications.

a. Projectile extraction. The force necessary to extract the projectile from the cartridge case shall be 1100 to 2800 pounds, inclusive (4.5.1.1).

b. Projectile torque. The projectile shall withstand a minimum torque of 10 inch pounds without movement (4.5.1.2).

3.3 Contamination of energetic components. Extreme care shall be exercised to avoid contamination of primers and propellant by oil, grease or other foreign matter since such contamination can cause misfires, increased action time and lot rejection.

3.4 Functioning. The cartridge shall comply with the following:

3.4.1 Conditioned at 68°F to 72°F.

a. Velocity. The average velocity of the projectile shall be 2680 $\pm$ 50 feet per second (fps) at 78 feet from the muzzle of the gun and the standard deviation of the velocity shall not exceed 40 fps. (see 6.7).

b. Pressure. The average chamber pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) (see 6.8) of the chamber pressure shall not exceed 51,000 pounds per square inch (psi).

c. Waterproofness. The average velocity (wet) of projectiles shall not vary from the average velocity (dry) of projectiles by more than 100 fps.

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3.4.2 Conditioned at ambient temperature.

a. Function and casualty. Cartridges shall function without casualty when fired in the automatic weapons.

b. Projectile plate function. Projectile shall function with high order detonation upon target impact. (Chapter 1, Section 6, AMCR 715-505, Vol. 8 and See 6.14 and 6.15).

c. Accuracy. The average mean radius of all targets shall not exceed 15 inches (accuracy) at a range of 600 yards.

3.4.3 Conditioned at -65°F.

a. Action time. The action time of the sample cartridges and the average action time ($\overline{X}$) plus four sample standard deviations shall not exceed 4.0 milliseconds (see 6.7).

3.4.4 Safety. There shall be no premature burst (see 6.16) or evidence thereof nor any projectile sticking in the gun bore in any of the test firings.

3.5 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3 and 6.13).

3.6 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.6.1 Processing defects. The metal parts and assemblies shall be free of splits, perforations, scratches (other than draw scratches), folds wrinkles, buckles, bulges, dents, cracks, rupture, deformaties and other forms of substandard workmanship.

3.6.2 Cleanliness. All components and assemblies shall be free of foreign matter, such as oil, grease, corrosion, dirt, chips, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawing	Quantity
Charged Body, Projectile	7259065	125
Projectile Assembly	7259064	125
Cartridge	7259105	430
Packaging	(see 5.1)	7

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein will result in sample disapproval.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-50388A (AR)

PARAGRAPH	TITLE: Cartridge, 20mm, HEI, M210 1/ and Components		SHEET 1 OF 2		DRAWING NUMBER: SEE BELOW; NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Charged Body, Projectile (Dwg. 7259065)				
	Examination for defects	125	Acc-0 Rej-1	3.2	4.4.2.1
	Projectile Assembly (Dwg. 7259064)				
	Examination for defects	125	Acc-0 Rej-1	3.2	4.4.2.2
	Torque test of fuze	125	Acc-0 Rej-1	3.2	Gage/4.5.1.4 5/
	Cartridge - (Dwg. 7259105)				
	Examination for defects	430	Acc-0 Rej-1	3.2	4.4.2.3
	Test Requirements				4.5.1
	Non functioning				
	(1) Projectile extraction	20	Acc-0 Rej-1	3.2a	4.5.1.1/11075131/ 7259443 2/
	(2) Projectile torque	20	Acc-0 Rej-1	3.2b	4.5.1.2/7259469
	(3) Propellant contamination/charge weight verification	20	Acc-0 Rej-1	3.3	4.5.1.3 2/

NOTES: 1/ MIL-STD-651 and dwg. 7553663 shall apply in defining and evaluating cartridge visual defects.
2/ Test may be conducted using same sample cartridges.
3/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the first article quantity, second sample not permitted.
4/ Velocity and pressure readings shall be recorded in a single test using the same sample cartridges.
5/ When the fuze turns with respect to the projectile body upon application of the specified torque in a counter clockwise direction, the cartridge shall be classified as defective.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50388A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
	Cartridge, 20mm, TP, M204 1/		7259105
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
				3.4	
	Functioning			3.4.1	4.5.2/4.5.2.1
	(1) Conditioned at 68°F to 72°F			3.4.1a	4.5.2.2 4/
	(a) Velocity	20	3/	3.4.1b	4.5.2.2 4/
	(b) Pressure	20	3/	3.4.1c	4.5.2.2
	(c) Waterproofness	20	3/		
				3.4.2	4.5.3
	(2) Conditioned at ambient			3.4.2a	4.5.3.1
	(a) Function and casualty	200	Acc-0 Rej-1		
	(b) Projectile plate function	50	Acc-0 Rej-1	3.4.2b	4.5.3.2
	(c) Accuracy	50	3/	3.4.2c	4.5.3.3
				3.4.3	4.5.4/4.5.4.1
	(3) Conditioned at -65°F			3.4.3a	4.5.4.2
	(a) Action time	50	3/		
	Packaging (containers)				
	Examination for defects	7	Acc-0 Rej-1	3.1a	4.4.4/5.1

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain.

a. Cartridge cases from one interfix lot number from one manufacturer.

b. Projectiles from one interfix lot number from one manufacturer.

c. Fuze assemblies from one interfix lot number from one manufacturer.

d. Primers from one interfix lot number from one manufacturer.

e. Propellant from one lot number from one manufacturer.

f. High explosive from one interfix lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Major and minor defects. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.5 percent and the AQL for minor defects shall be 2.5 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, when three or less defects appear in a classification, sampling plans applicable to a class shall not be used for major defects.

c. Critical defects. Unless otherwise specified, one hundred percent examination shall be performed for all critical defects. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot shall then be screened and resubmitted for visual inspection of critical defects. If a critical defect is found during packing, the portion of the lot that has been packed or is in the process of being packed shall be rejected. In addition the portion of the lot remaining to be packed shall be rejected. The lot shall then

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be screened and resubmitted for visual inspection of critical defects.

d. Cartridge. Each cartridge shall be inspected in process for chamber gage failure (profile and alignment, max.). Cartridges which fail to meet the requirement of 3.2 shall be removed.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50388A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Charged Body, Projectile	1 OF 1	7259065
			NEXT HIGHER ASSEMBLY: 7259064

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Explosive on threads at indicated length		100%	3.2	Visual
2.	Depth, mouth to HE mix		100%	3.2	Gage
Major					
101.	Bourrelet diameter, max		.25%	3.2	8655592
Minor					
201.	Poor workmanship		.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50388A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Projectile Assembly	1 OF 1	7259064
			NEXT HIGHER ASSEMBLY: 7259105

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Disassembly torque test of fuze		.25%	3.2	Gage/4.5.1.4 1/ 2/
Minor					
201.	Gap, maximum, between fuze and body		.40%	3.2	11075013 1/
202.	Poor workmanship		.65%	3.6	Visual

NOTES: 1/ May be performed at cartridge acceptance.
2/ When the fuze turns with respect to the projectile body upon application of the specified torque in a counter clockwise direction, the projectile assembly shall be classed defective.

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50388A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Cartridge, 20mm, HEI, M210 1/ (examinations)				7259105
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Split or perforated case		100%	3.2	Visual
2.	Crack or split in projectile		100%	3.2	Visual
3.	Case to projectile crimp missing		100%	3.2	Visual
Major					
101.	Chamber gage failure (profile & alignment, max)	300	Acc-0 Rej-1	3.2	7299942/7286068 7286069/7286070/ 7286071 2/
102.	Depth of primer seating		.25	3.2	7569084
103.	Total length, max		.25	3.2	7556901
104.	Primer cocked, loose, missing or inverted		.25	3.2	Visual/Manual
105.	Corrosion		.25	3.2	Visual
106.	Head Configuration		.25	3.2	Visual
107.	Improper or incomplete crimp (case-projectile)		.25	3.2	Visual
108.	Wax coating missing or incomplete		.25	3.2	Visual
109.	Fuze nose cap loose		.25	3.2	Visual/Manual

NOTES: 1/ MIL-STD-651 and 7553663 shall apply in defining and evaluating cartridge visual defects. 2/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50388A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	Cartridge, 20mm, HEI, M210 1/ (examinations)				7259105
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Waterproofing missing around primer		.40	3.2	Visual
202.	Primer crimp missing or incomplete		.40	3.2	Visual
203.	Rotating band damaged		.40	3.2	Visual
204.	Foreign matter except corrosion		.40	3.2	Visual
205.	Projectile protective coating damaged or incomplete		.40	3.2	Visual
206.	Marking incorrect, incomplete, illegible or missing		.40	3.2	Visual
207.	Cartridge case damaged		.40	3.2	Visual
208.	Scratches (other than draw) folds, wrinkles, buckles, bulges, dents		.40	3.2	Visual
209.	Wax coating on projectile		.40	3.2	Visual
210.	Wax coating flaky or otherwise unsatisfactory		.40	3.2	Visual
211.	Poor workmanship		.65	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50388A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cartridge, 20mm, HEI, M210 (Tests) 1/	1 OF 1	7259105
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Tests				
	Non-functioning				
	(1) Projectile extraction	20	4.4.3.1	3.2.a	4.5.1 4.5.1.1 11075131/7259443 2/
	(2) Projectile torque	20	4.4.3.2	3.2.b	4.5.1.2/7259469
	(3) Propellant contamination/charge weight verification	20	Acc-0 Rej-1	3.3	4.5.1.3 2/
	Functioning			3.4	
	(1) Conditioned at 68°F to 72°F		4.4.3.4	3.4.1	4.5.2/4.5.2.1
	a. Velocity	20	4.4.3.4 a	3.4.1.a	4.5.2.2 3/
	b. Pressure	20	4.4.3.4 b	3.4.1.b	4.5.2.2 3/
	c. Waterproofness	20	4.4.3.4 c	3.4.1.c	4.5.2.2
	(2) Conditioned at Ambient			3.4.2	4.5.3
	a. Function & Casualty	200	4.4.3.5	3.4.2.a	4.5.3.1
	b. Projectile Plate Function	50	4.4.3.7	3.4.3.b	4.5.3.2
	c. Accuracy	50	4.4.3.6	3.4.2.c	4.5.3.3
	(3) Conditioned at -65°F			3.4.3	4.5.4/4.5.4.1
	a. Action Time	50	4/	3.4.3.a	4.5.4.2

NOTES: 1/ MIL-STD-651 and 7553663 shall apply in defining and evaluating cartridge visual defects.

2/ Test may be conducted using same sample cartridges.

3/ Velocity and pressure readings shall be recorded in a single test using the same sample cartridges.

4/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot.

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4.4.3 Tests. The tests listed in 4.4.2.4 shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only cartridges which have met the visual and dimensional requirements and which have been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be mixed thoroughly before being divided into samples for the various tests.

4.4.3.1 Projectile extraction.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall be cause for rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined sample fail to comply with the maximum requirement.

4.4.3.2 Projectile torque. Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the applicable requirement.

4.4.3.3 All cartridge firing tests. The lot shall be rejected if in any firing test one or more of the following casualties occur:

a. Misfire associated with c or d of paragraph 6.10.

b. Projectile remaining in bore.

c. Metal parts separation, except rotating band separation (see 6.11.1).

d. Premature projectile functioning (see 6.16)

4.4.3.4 Cartridge functioning at 68°F to 72°F. If any requirement(s) in paragraph 3.4.1 is not achieved on the first sample, a second sample shall be tested for the failed parameter(s).

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a. Velocity. If the sample fails to comply with either or both requirements a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the retest samples fails tocomply with the applicable requirement(s). (See 6.7)

b. Pressure.If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 51,000 psi a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to comply with the applicable requirements. (See 6.8).

c. Waterproofness. If the average velocity (wet) of sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirements of 3.4.1.a.

4.4.3.5 Function and casualty. Two-hundred cartridges shall be subjected to this test. The lot shall be rejected if, at any time, any of the following occurs:

a. Any fire "out of battery" (see 6.11.2).

b. Any primer misfires (see 6.10).

c. Any cartridge case fails to extract.

d. Any cartridge case longitudinal split occurs (L or M) (Dwg. 7553663).

e. Any cartridge case circumferential rupture, partial (L) or complete (Dwg. 7553663).

f. Any detached metal from cartridge case occurs (see 6.11.3).

g. Any cartridge fails to chamber.

h. Projectile remaining in bore.

In addition, if defects listed in Table II, occur equal to or less than the "ACC" number, the lot shall be accepted. If any defect(s) occur equal to or exceeding the "REJ" number, the lot shall be rejected. If any defect(s) occur in excess of the "ACC" number, but less than the "REJ" number, a second sample of 400 cartridges shall be selected and fired in the M3 or MK16 weapon. The lot shall be rejected if any defect occurs equal to or greater than the "CUMULATIVE REJ" number.

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TABLE II. Firing Defects

		ACC	REJ	CUMULATIVE REJ
1.	Primer missing	0	2	2
2.	Primer leak (see Note 1)	9	20	25
3.	Longitudinal case split (I or S) (Dwg. 7553663)	10	21	31
4.	Longitudinal case split (K) (Dwg. 7553663)	0	2	2
5.	Case circumferential rupture, partial (S, J or K) (Dwg. 7553663)	4	9	11
6.	Longitudinal case split (J) (Dwg. 7553663)	4	9	11
7.	Projectile rotating band separation complete or partial separation having band seat serration imprint (see 6.11.4)	0	3	3
8.	Projectile rotating band separation having greater than 0.5 inch and without band seat serration imprint (see 6.11.4)	1	4	6

Note 1. For primer defect definitions, see Chapter 1, Section 3, Paragraph 3.6 of AMCR 715-505, Volume 8.

4.4.3.6 Accuracy. If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of 10 retest accuracy targets exceeds the applicable requirement (see 3.4.2.b).

4.4.3.7 Projectile plate function. Fifty cartridges shall be subjected to this test. The lot shall be rejected if five or more projectiles fail to detonate high order upon impact. The lot shall be accepted if not more than two projectiles fail. If more than two or less than five projectiles fail, an additional fifty cartridges shall be tested for the projectile plate function parameter only. If the cumulative number of failures equals or exceeds 7, the lot will be rejected.

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4.4.4 Packaging, packing and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644.

4.4.5 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.12 herein.

4.4.5.1 Ballistic test equipment. The ballistic test equipment required to perform the tests prescribed herein, shall be as specified in AMCR 715-505 and by the design activity.

4.5 Methods of inspection.

4.5.1 Non-functioning.

4.5.1.1 Projectile extraction. The cartridge shall be placed in the applicable fixture specified herein and the tension force specified on the applicable drawing shall be applied. Projectile must separate from the cartridge case in the range specified on the applicable drawing. The method of test shall be as specified in Chapter 1, Section 10 of AMCR 715-505, Volume 8. This is a destructive test.

4.5.1.2 Projectile torque. The cartridge shall be marked with a light scratch extending axially across the rotating band onto the case neck. Then the cartridge shall be placed in the applicable fixture specified herein and the inch pound torque specified on the applicable drawing shall be applied slowly to the projectile. Observation shall be made for movement of the projectile with respect to the misalignment in the scratch mark. This test is a non-destructive test.

4.5.1.3 Propellant contamination and charge weight verification. After completion of test specified in 4.5.1.1 each of the cartridges tested shall be examined visually for contamination as described in Chapter 1, Section 10, of AMCR 715-505, Volume 8. The charge from each of the sample cartridges shall be weighed and recorded.

4.5.1.4 Disassembly torque test of fuze on projectile. After six (6) hours curing time, the projectile shall be placed in the applicable fixture and the inch pound requirement specified on the applicable drawing shall be applied. Observation shall be made for the requirement specified on the applicable drawing. This test is a non-destructive test.

4.5.2 Functioning (conditioned at 68°F to 72°F).

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4.5.2.1 Ammunition conditioning. The cartridges shall be conditioned for a period of not less than six (6) hours after the conditioning chamber has stabilized at 68°F to 72°F. Timing for the conditioning period shall start when the chamber has stabilized after the sample has been placed in the conditioning chamber.

4.5.2.2 Velocity, pressure and waterproofness. These tests shall be conducted in accordance with Chapter 1, Section 2 of AMCR 715-505, Volume 8. The sample standard deviation shall be computed in accordance with 6.7 and 6.8 respectively.

4.5.3 Functioning (conditioned at ambient temperature).

4.5.3.1 Function and casualty. The method of testing shall be as specified in Chapter 1, Section 3 of AMCR-715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges in the M3 or MK16 gun. The weapon barrel shall be at ambient temperature at the beginning of the test and cooled to ambient after each 100 rounds. There shall be observation for a minimum distance of 400 feet from the gun muzzle for premature burst.

4.5.3.2 Projectile plate function. This test shall be conducted in accordance with AMCR 715-505, Chapter 1, Section 6, Volume 8. The aluminum target (2024 T4, T3 or T36) of .040 inch nominal thickness shall be located 200 ± 25 yards from the muzzle of the gun normal to the line of fire. Observation for projectile functioning shall be made from camera position.

4.5.3.3 Accuracy. The method of test and measurement of targets shall be in accordance with Chapter 1, Section 4 of AMCR-715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards, the average mean radius shall be reduced in direct proportion to the reduction in range. The 200 yard range may be reduced by 11.5 yards to accommodate combining the accuracy and projectile plate functioning tests. Any variation from the 200 yard range requires direct proportional adjustment of the average mean radius (see 6.15).

4.5.4 Functioning (conditioned at -65°F).

4.5.4.1 Ammunition conditioning. The cartridges shall be conditioned for a period of not less than six (6) hours after the conditioning chamber has stabilized at -65°F ± 5°F. Timing for the conditioning period shall start when the chamber has stabilized after the sample has been placed in the conditioning chamber.

4.5.4.2 Action time. The method of testing shall be in accordance with Chapter 1, Section 5, of AMCR 715-505, Volume 8.

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4.5.5 Defective weapon. If a test weapon fails to comply with the minimum cyclic rate during a normal burst operation or its found to be defective during or at the end of burst, the test shall be suspended and Government approval shall be required for a new test. The cyclic rate of each burst shall not be less than 680 shots per minute in the M3 or MK16 gun. The cyclic rate for each burst shall be recorded.

4.5.6 Test validity. If for any reason the test activity considers that the test conditions have detrimentally affected the test results, the test activity may request the Government to declare the test invalid and authorize a new test.

5. PACKAGING

5.1 Level A.

5.2 Packing. The M210 Cartridge shall be packed in accordance with instructions specified by the procuring activity.

5.3 Marking. Marking of the packed container shall be in accordance with MIL-STD-129, Code of Federal Regulations Title 49, Parts 100-199.

6. NOTES

6.1 Intended use. This specification covers the loading assembling and packing for Cartridge, 20mm, High Explosive Incendiary, M210 for use in M3 or MK16 weapons designed to utilize percussion primed cartridges with the M21 type cases.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by AMCCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, ARDEC, ATTN: AMSMC-QAF-S(D) and SMCAR-CCL-CE, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Safety data. For the safety data requirements refer to Safety Data Document as listed on Product Drawings.

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6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, U.S. Army Armament Research and Development Command (ARRADCOM) or U.S. Army Armament Reseach and Development Center (ARDC) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.6 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.7 Standard deviation. Standard deviation shall be calculated from the following formula:

$$S = \sqrt{\Sigma(X_i - \bar{X})^2/(n-1)}$$

where:

X_i = each individual value

$\bar{X}$ = sample arithmetic mean = $\Sigma X_i/n$

n = sample size

$\Sigma(X_i - X)^2$ = The sum of the squares of the differences between the sample mean and each individual value

6.8 Range plan. Where a range plan is specified, the individual values will be recorded in subgroups of 5 in the order in which they are observed. The ranges for the subgroups shall be added and the sum of the ranges shall be divided by the number of subgroups to determine the average range ($\bar{R}$).

6.9 Intermediate point inspection. The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the end item.

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6.10 Misfires. Misfired test cartridges, except those from the waterproofness test shall be retested in equipment identified in the accuracy test. If a cartridge fails to function in that test, examinations shall be conducted for the following conditions:

a. Excessive primer seating depth.

b. Missing or obstructed case vent.

c. Missing or insufficient primer composition or propellant.

d. Partially burned or contaminated primer compostion or propellant.

The lot shall not be penalized if condition a. is observed. However, this condition shall be reported for information. The "ACC" number shall be zero (0) and the "REJ" number 3, for misfires associated with condition b. The second sample for evaluation of misfires associated with condition b, shall consist of the function and casualty second sample (see 4.4.3.5). The lot shall be rejected if either condition c or d is observed. Misfire(s) occurring in the waterproofness test shall be handled in accordance with AMCR 715-505, Volume 8.

6.11 Defect definitions.

6.11.1 Metal part separation. Separation or breakup of projectile part(s), except rotating band separation, as evidenced by recovery of the part(s) or fragment(s), or hole(s) in the fragmentation screen(s).

6.11.2 "Out-of-battery". A cartridge shall be classed defective if it fires, due to delayed ignition, after the bolt has been unlocked and extraction has been initiated or completed.

6.11.3 Detached metal from cartridge case. Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears. Shavings of metal from the interior wall of the case neck, in the crimped area shall not be considered as detached metal.

6.11.4 Rotating band separation. For classification as a defect, there must be evidence by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). Normal band fringing shall not be classed as a defect.

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6.12 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000.

6.13 First article sample. In order to minimize inspection costs, the firing test of 4.4 shall be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the test facility.

6.14 Projectile function. The photographic standards shown in Chapter 1, Section 6, Para 6.6 of AMCR, 715-505, Volume 8 are applicable for comparison of the projectile performance for the HEI cartridge functioning test.

6.15 Testing precaution. If an open projectile impact area is in use at the test facility, precaution should be used because of the fire hazard.

6.16 Premature burst. A premature burst is considered to have occurred if the projectile functions in the gun bore or prior to impact.

6.17 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.18 Subject term (key word) listing.

High Explosive Incendiary
Small Arms Ammunition
20mm Automatic Weapon Systems

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AC99)

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MIL-C-50388A (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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